[ORIGINAL RESEARCH]

Prevention of Schizophrenia Relapse with Extended Release Quetiapine Fumarate Dosed Once Daily: A Randomized, Placebo-Controlled Trial in Clinically Stable Patients

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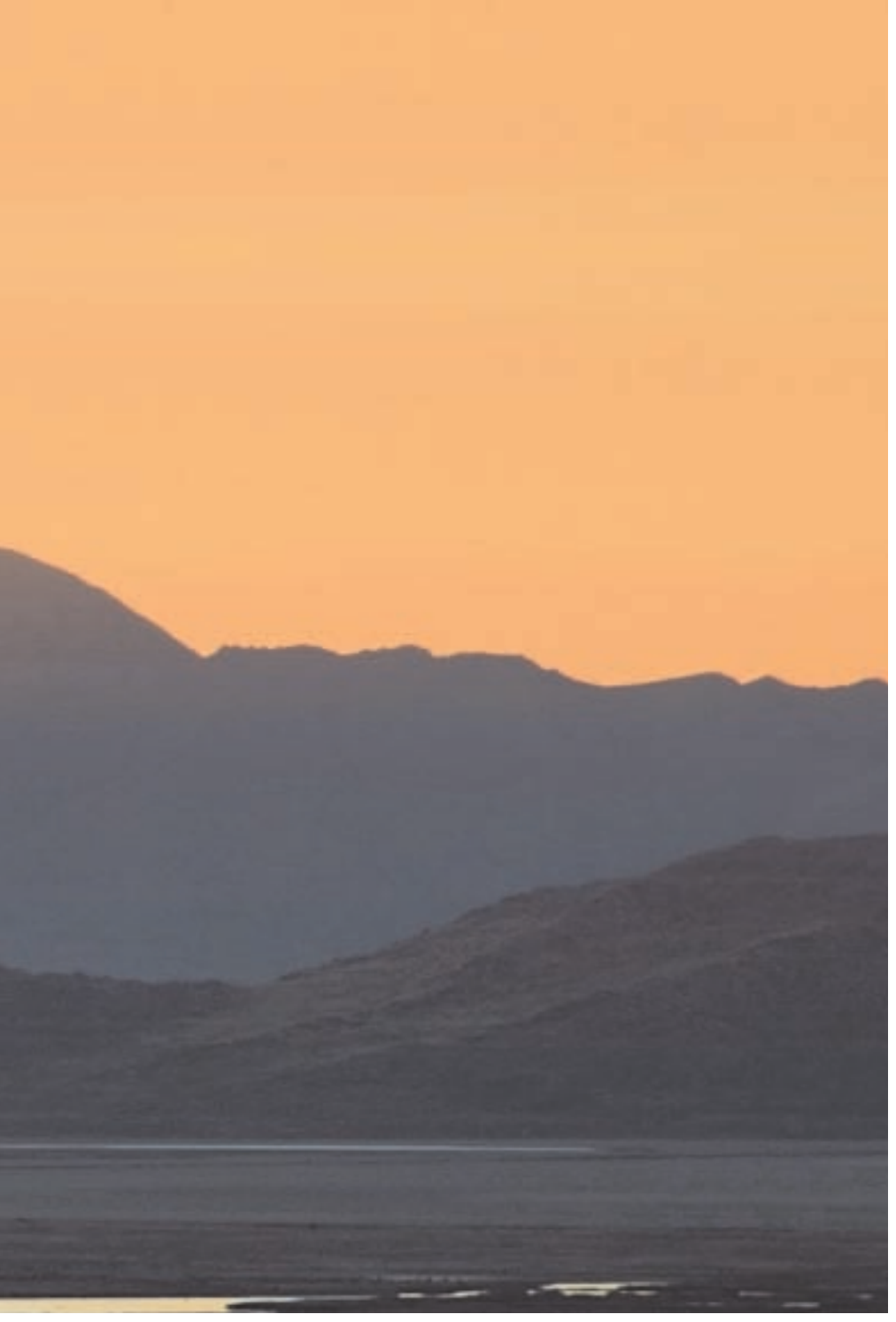
KEY WORDS: atypical antipsychotics, extended-release preparations, dose initiation, quetiapine, schizophrenia, relapse prevention

ABSTRACT

Introduction: This long-term, randomized, double-blind, placebo-controlled study examined the efficacy of extended release quetiapine fumarate (quetiapine XR) in preventing psychotic relapse in schizophrenia.

Methods: Three hundred twenty-seven clinically stable patients with schizophrenia were switched to open-label quetiapine XR (300mg on Day 1, 600mg on Day 2, followed by flexible dosing [400–800mg/day]) for a 16-week stabilization phase. Thereafter, patients who were clinically stable for four months were randomized to flexible doses of quetiapine XR (400–800mg/day) or placebo. Primary endpoint was time to first schizophrenia relapse after randomization. Secondary endpoints included risk of relapse at six months. Interim analyses were planned after 45 and 60 relapses and final analysis after 90 relapses. Maximal treatment time was one year.

Results: The study was terminated after the first interim analysis showed a significant difference between randomized treatment groups. Time to relapse was significantly longer in quetiapine XR-treated patients versus placebo (hazard ratio 0.16 [95% confidence interval 0.08, 0.34]; $p<0.001$). Fewer quetiapine XR-treated patients relapsed versus those receiving placebo (10.7% vs. 41.4%, respectively). Estimated risk of relapse at six



months was significantly lower with quetiapine XR (14.3%) compared with placebo (68.2%; $p<0.0001$). The incidence of treatment-related adverse events (AEs) was similar between quetiapine XR and placebo groups (18% and 21% of patients, respectively) and only one percent of patients in each group withdrew because of AEs.

Conclusion: Once-daily quetiapine XR (400–800mg/day) was effective in preventing relapse in patients with clinically stable schizophrenia. Quetiapine XR was well tolerated during longer-term use.

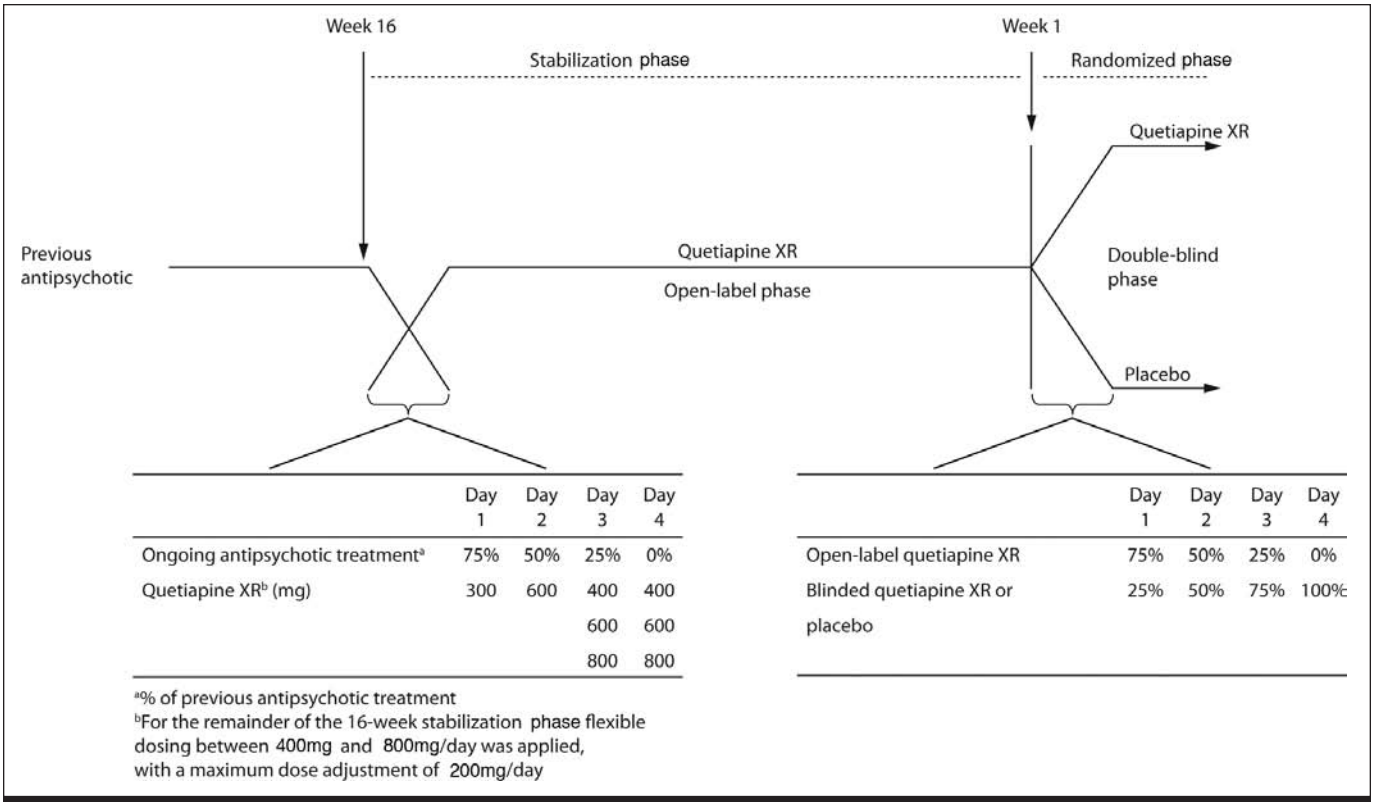
INTRODUCTION

Schizophrenia is a chronic disorder characterized by alternating periods of acute exacerbation and full or partial remission. While atypical antipsychotics have proven efficacy in reducing acute symptoms and preventing or delaying relapse in some studies, relapse rates are still high,[1] resulting in an increased economic burden[2] and poor prognosis.[3] Since the main objective of antipsychotic treatment (aside from the initial management of symptoms) is to prevent relapse without increasing the potential for adverse effects, this indicates a continuing unmet need in the treatment of schizophrenia.

Predictors of relapse include poor adherence to treatment, severe residual psychopathology, lack of insight into illness, comorbid substance abuse, and inadequate relationships with family and care providers.[4] Of these, poor or nonadherence to treatment is often the major contributory cause.[5–7] This can be a result of many factors related to the patient (psychopathology, cognitive impairment, age, comorbidity, gender, personality traits, insight), the treatment (tolerability, route of administration, pattern and complexity of dosing, length of treatment, cost of treatment, polypharmacy, onset of action, efficacy), and the treatment context (therapeutic relationship, social support, attitude of both patient and physician toward treatment, supervision of treatment, social rank of illness, location of treatment provision).[8]

When compared with conventional antipsychotics, the lower levels of extrapyramidal symptoms (EPS) and adverse events (AEs) generally experienced by patients prescribed atypical antipsychotics can be expected to result in improved adherence to treatment and, consequently, lower levels of relapse.[9] Indeed, a meta-analysis of patients with schizophrenia has shown significantly lower relapse rates in patients treated with atypical antipsychotics compared with those treated with haloperidol.[1]

The atypical antipsychotic quetiapine is a first-line treatment for schizophrenia. It has



	Day 1	Day 2	Day 3	Day 4
Ongoing antipsychotic treatment[a]	75%	50%	25%	0%
Quetiapine XR[b] (mg)	300	600	400	400
			600	600
			800	800

	Day 1	Day 2	Day 3	Day 4
Open-label quetiapine XR	75%	50%	25%	0%
Blinded quetiapine XR or placebo	25%	50%	75%	100%

[a]% of previous antipsychotic treatment
[b]For the remainder of the 16-week stabilization phase flexible dosing between 400mg and 800mg/day was applied, with a maximum dose adjustment of 200mg/day

FIGURE 1. Study design

demonstrated efficacy across a broad range of symptoms and is comparable in this respect with other atypical antipsychotics.[10] Quetiapine has shown efficacy in treating anxiety symptoms in bipolar disorder[11] and agitation in schizophrenia.[12] In addition, early evidence suggests efficacy in treating major depressive disorder and generalized anxiety disorder.[13–15] Moreover, quetiapine has a lower propensity to cause EPS than chlorpromazine,[16] haloperidol,[17] or risperidone,[18] and has an incidence of EPS (including akathisia) and prolactin levels similar to placebo across the dose range[19] in patients with schizophrenia. Quetiapine has also been shown to be associated with a lower risk of tardive dyskinesia in patients with schizophrenia compared with conventional antipsychotics.[20]

Quetiapine is available as an immediate release (IR) formulation and has shown efficacy through twice- or three-times-daily administration for schizophrenia and twice-daily administration for bipolar mania. Quetiapine IR is also approved for once-daily administration for bipolar depression in the USA. Extended release quetiapine fumarate (quetiapine XR) is a new, once-daily formulation based on a gel matrix technology to control the release of quetiapine. It relies on delayed drug release from the formulation to maintain plasma drug concentrations at higher levels for a longer time period than is possible with an IR formulation. The extended-release characteristics of the quetiapine XR formulation result in less frequent dosing being required to maintain therapeutic drug concentrations. Peak plasma quetiapine concentrations (C_{max}) occur approximately six hours after administration (t_{max}), compared with approximately one hour for the IR formulation. Quetiapine XR has a predictable and reproducible pharmacokinetic profile and is formulated to be administered once daily. It has been developed to provide patients and physicians with a more convenient dosage and a simpler dose-administration regimen and has recently been approved by the US Food and Drug Administration (FDA) in 2007 for the treatment of acute schizophrenia. A therapeutically effective dose should be achieved with this formulation by Day 2 without compromising the safety and tolerability profile of the treatment.

A randomized, placebo-controlled study of quetiapine XR has demonstrated that once-daily quetiapine XR is effective and well tolerated in patients with acute schizophrenia.[21] Given the importance of adherence to treatment in achieving successful long-term therapy, it is possible that the new XR formulation of quetiapine may be appropriate and efficacious in maintenance treatment. Thus, the objective of this study was to assess the long-term efficacy of quetiapine XR by evaluating relapse prevention in patients with chronic

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schizophrenia. This study employed a unique design by recruiting clinically stable patients with minimal symptomatology. These patients were then assessed for clinical stability over four months before entering the randomization phase. The study design also included monitoring by an external Data and Safety Monitoring Board (DSMB) in order to address ethical considerations and to minimize the potential risk to patients receiving placebo treatment.

MATERIALS AND METHODS

Patients. Patients from centers in Bulgaria, India, Poland, Russia, and Ukraine were considered eligible for inclusion in the study if they were ages ≥18 to ≤65 years; had a documented clinical diagnosis of schizophrenia (according to *Diagnostic and Statistical Manual for Mental Disorders, Fourth Edition* [DSM-IV]) for at least two years; were clinically stable before entering the stabilization phase (defined as a Clinical Global Impression-Severity of Illness [CGI-S] score ≤4 and unchanged treatment [both compound and dose] with antipsychotic agent[s] within four weeks prior to entering the study); and had a Positive and Negative Syndrome Scale (PANSS) total score ≤60 at enrollment (Week 16). Exclusion criteria included treatment with depot antipsychotics within one dosing interval before enrollment (Week 16); pregnancy or breastfeeding; any DSM-IV Axis 1 disorder not defined in the inclusion criteria; any clinically significant deviations from the reference range in clinical laboratory test results at enrollment, as evaluated by the investigator; intolerance or lack of response to quetiapine; previous treatment with clozapine and/or valproic acid within two months of enrollment; and history of nonadherence, as judged by the investigator.

The study protocol was approved by the relevant local ethics committees and written informed consent was provided by each patient (or the patient's legally authorized representative) prior to the initiation of any study-related procedures.

Study design. This was a one-year, multicenter, randomized, double-blind, parallel group, placebo-controlled study (number D1444C00004) in patients with clinically stable schizophrenia. Patients were assessed over a 16-week stabilization phase prior to randomization during which they were switched from their current antipsychotic to open-label quetiapine XR. Switching was achieved using cross-titration, with quetiapine XR dosed in the evening at 300mg on Day 1, 600mg on Day 2, then flexibly dosed at 400, 600, or 800mg/day from Day 3 onward (Figure 1). Previous antipsychotic medication was tapered to 75 percent on Day 1, 50 percent on Day 2, 25 percent on Day 3, and stopped at Day 4.

At the end of the 16-week, open-label stabilization phase, patients who were on an established dose of quetiapine (range 400–800mg/day; dose options 400, 600, or 800mg/day), had a CGI-S score ≤4 and a PANSS total score of ≤60 at -16 weeks (enrollment), with no change ≥10 points in PANSS total score from -16 weeks to -8 weeks and from -16 weeks to baseline (randomization), were assessed as clinically stable and allowed to enter the randomized, double-blind phase.

The double-blind portion of the study started at randomization (defined as study baseline), at which point the patients were assigned to treatment with either quetiapine XR (flexibly dosed at 400–800mg/day) or placebo. Following randomization, a cross-titration of four days started whereby the open-label quetiapine XR was phased out and blinded quetiapine XR (or placebo) was phased in (Figure 1). Treatment was planned to last for one year or until relapse.

For safety and ethical reasons, interim analyses were planned after 45 and 60 observed relapses, allowing for termination of the study if the primary endpoint showed a significant difference between treatments (based on a requirement of $p<0.004455$). An independent DSMB external to the sponsor performed the interim analyses. Final analysis would be performed after 90 relapses.

Outcome measures. *Primary endpoint.* The primary endpoint was the time to first schizophrenia relapse after randomization. Relapse was defined as at least one of the following: hospitalization due to worsening schizophrenia, increase in PANSS score of ≥30 percent from baseline, Clinical Global Impression-Improvement (CGI-I) score ≥6 (much worse or very much worse), or a need for additional antipsychotic medication to treat psychosis (as determined by the investigator).

Secondary endpoints. The efficacy of quetiapine XR versus placebo was also assessed by estimating the relapse rate at six months. In addition, PANSS total score and PANSS positive, negative, and general psychopathology subscale scores were assessed monthly. The proportion of patients with CGI-S score ≤4 at the last visit, the proportion of patients with CGI-I score ≤4 at the last visit, and the change in CGI-S score from randomization to the last visit were also assessed.

The safety and tolerability of quetiapine XR were assessed by monitoring the number and severity of reported AEs and withdrawals during the stabilization and double-blind randomization phases. AEs were also monitored during the first two weeks after enrollment. Laboratory measurements, including hematology, clinical chemistry (P-glucose, S-insulin, and HbA_{1c}), lipids, thyroid function, and urinalysis, were made at enrollment, every four weeks during the stabilization phase (excluding urinalysis), and at

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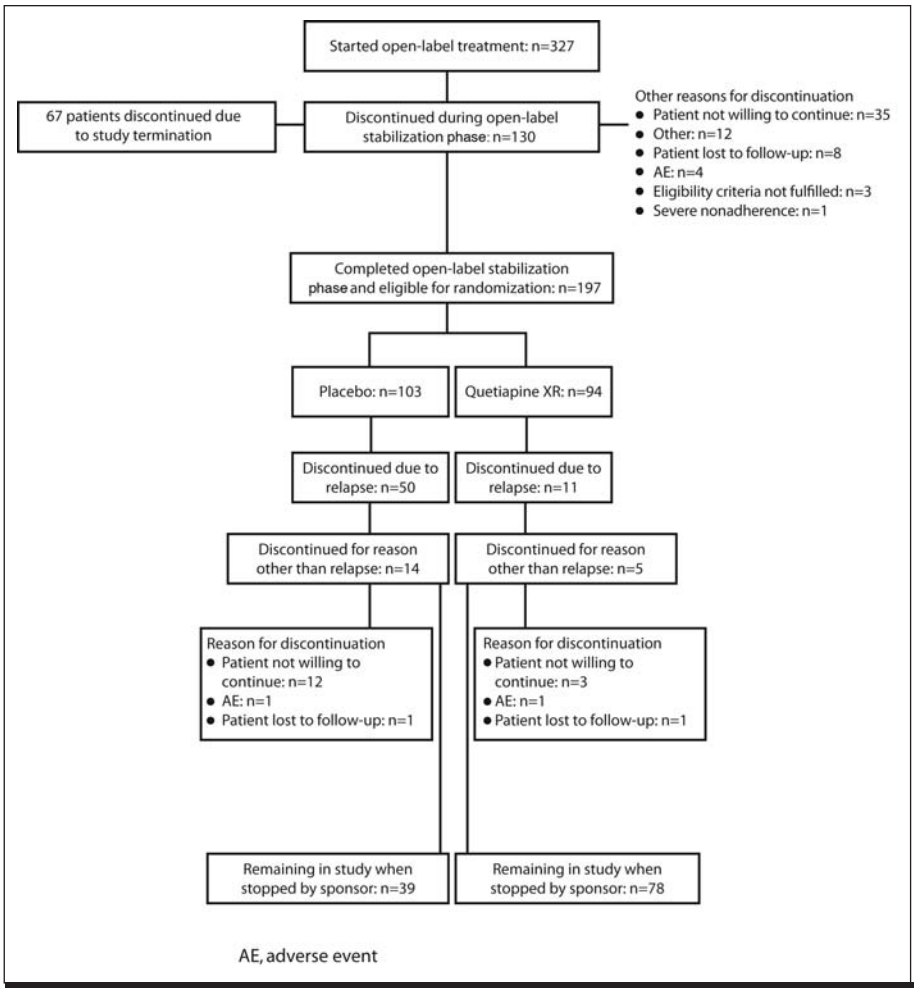


FIGURE 2. Patient disposition

baseline, Month 3, Month 6, Month 9, and Month 12 of the randomization phase. Although laboratory measurements were recorded for all patients, fasting glucose, HbA_{1c}, insulin, and lipids values from patients with a fasting time of at least eight hours since the last meal documented in their case report forms are reported here. An electrocardiogram was performed at enrollment, at randomization, Month 6, and Month 12. Additionally, measurements of vital signs (blood pressure and pulse rate) were performed at enrollment, monthly during the stabilization phase, one month after randomization, and then followed by quarterly assessments. Finally, changes in body weight and waist circumference were measured at enrollment, monthly during the stabilization phase, one month after randomization, and then followed by quarterly assessments.

The incidence and severity of EPS were assessed by measuring change in the Simpson-Angus Scale (SAS) and Barnes Akathisia Rating Scale (BARS) every two weeks from enrollment to baseline and every month throughout the randomized phase. The use of anticholinergic medication and reporting of EPS-related AEs were also monitored.

Additional endpoints. Adherence was assessed by performing a tablet count at each visit and was calculated based on the difference between number of dispensed and returned tablets. Fully adherent patients were classified as those who took between ≥70 percent and ≤120 percent of their doses.

Statistical analysis. The primary endpoint of time to first schizophrenia relapse was analyzed using a Cox proportional hazards model to estimate the hazard ratio (HR) of relapse rate between treatment groups. The time to event was censored when a patient left the study for reasons other than relapse or study completion.

The proportion of patients having a relapse at six months in each treatment group was estimated from the same Cox proportional hazards model, where one minus the estimation of the survivor function at six months provided the estimation for this proportion and the t-test was used to test the difference between treatment groups.

A robust five-step sequential procedure was used for the confirmatory part of this study to ensure a multiple level of significance of 0.05. The significance tests were performed sequentially in the following order:

1. Time to first schizophrenia relapse
2. Relapse rate at Month 6
3. PANSS total score over time
4. CGI-S score ≤4 at final visit
5. Mean CGI-I score at final visit.

A two-sided test for each of these null hypotheses was performed. The significance level of the first test (primary analysis of interim intent-to-treat [ITT] population) was adjusted to account for the interim analyses using the O'Brien-Flemming boundaries, i.e. α=0.004455. For the subsequent tests (tests 2–5), α=0.05 was used. A successful outcome of the first test in the stepwise analysis was defined as a HR that was statistically significantly lower than 1 when comparing quetiapine XR with placebo.

No confirmatory claims for the above tests were made unless the null hypotheses in the preceding tests and in the test itself were all rejected. For example, no confirmatory claim for the test for the PANSS total score (Test 3) was made unless the null hypotheses for the tests of time to first relapse (Test 1), proportion of patients

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with relapse (Test 2), and PANSS total score (Test 3) were all rejected.

The stepwise sequential procedure ensures a multiple level of significance of 0.05 and concords to the Committee for Proprietary Medicinal Products guidelines.[22] No confirmatory claims were made for any other secondary outcome variables other than the ones stated above, but *p*-values are presented to aid in the interpretation of the results. The confidence levels and *p*-values presented are unadjusted for multiplicity.

The PANSS total and subscale scores were analyzed using a mixed effect repeated measurement analysis of all post-baseline measurements from randomization up to, but not including, the relapse visit. PANSS total and subscale scores were also analyzed from randomization up to and including the relapse visit. The difference between the treatment groups was estimated with the least squares means (LSM) method. A two-sided 95-percent confidence interval for this difference was calculated. The null hypothesis that there was no difference between the treatment groups was tested with a two-sided test of level 0.05. Score at randomization, treatment, and visit were included as fixed effects, and subjects nested within a treatment were treated as a random effect. This analysis adjusted for differences in baseline and variance between treatment groups by presenting the LSM.

Descriptive statistics were used for safety and tolerability assessments.

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RESULTS

Patients. A total of 327 patients were enrolled at 26 sites in five countries; 197 patients completed the open-label stabilization phase and were randomized. The majority of discontinuations during the stabilization phase were due to the early termination of the study (Figure 2). The interim analysis that was conducted after the first 45 relapses included 171 (87%) of the 197 randomized patients (87 in the placebo group and 84 in the quetiapine XR group). At the time of termination, another 26 patients had been randomized, resulting in a total of 197 randomized patients with 103 in the placebo group and 94 in the quetiapine XR group. When the study was stopped, 39 placebo-treated patients (37.9%) and 78 quetiapine XR-treated

TABLE 1. Patient baseline demographics and clinical details

	STABILIZATION PHASE	RANDOMIZED PHASE (INTERIM ITT POPULATION)	
	Quetiapine XR (*n*=327)	Placebo (*n*=87)	Quetiapine XR (*n*=84)
GENDER, *n* (%):			
Male	192 (59)	54 (62)	49 (58)
Female	135 (41)	33 (38)	35 (42)
AGE, YEARS, MEAN (RANGE)	36 (18–64)	33 (18–56)	37 (18–63)
DIAGNOSIS, YEARS, MEAN (RANGE):			
Age at first diagnosis	27 (6–51)	25 (13–51)	28 (14–48)
Years since diagnosis	9.3 (2–39)	8.3 (2–34)	9.1 (2–33)
Number of schizophrenic episodes	4.5 (1–34)	3.8 (1–17)	4.9 (1–32)
PSYCHOACTIVE MEDICATIONS AT ENROLLMENT, *n* (%):			
Any psychoactive medication	312 (95.4)		
Antipsychotics	310 (94.8)		
Conventional antipsychotic	154 (47.1)		
Atypical antipsychotic	176 (53.8)		
Olanzapine	47 (14.4)		
Risperidone	87 (26.6)		
Other	43 (13.1)		
>1 antipsychotic	49 (15.0)		
ETHNICITY, WHITE, *N* (%)	327 (100)	87 (100)	84 (100)
WEIGHT, KG, MEAN (RANGE)	70.9 (39–121)	73.9 (42–118)	73.0 (51–106)
PANSS SCORE, MEAN (SD)	53.4 (7.7)	48.1 (7.6)	48.3 (7.9)
CGI-S SCORE, MEAN (SD)	3.1 (0.8)	2.7 (0.8)	2.7 (0.7)

ITT, intent-to-treat; PANSS, Positive and Negative Syndrome Scale; CGI-S, Clinical Global Impression-Severity of Illness

TABLE 2. Summary of number (%) of patients reporting AEs over the entire study period

	STABILIZATION PHASE	RANDOMIZED PHASE	
	Quetiapine XR (*n*=327)	Placebo (*n*=103)	Quetiapine XR (*n*=94)
Total AEs*	163 (49.8)	42 (40.8)	30 (31.9)
Serious AEs*	0	2 (1.9)	0
Treatment-related AEs*†	133 (40.7)	22 (21.4)	17 (18.1)
AEs leading to discontinuation*	4 (1.2)	1 (1.0)	1 (1.1)
MOST COMMON AES (OCCURRING IN ≥5% OF PATIENTS)			
Somnolence*	63 (19.3)	-	-
Dizziness*	21 (6.4)	-	-
Insomnia*	-	18 (17.5)	8 (8.5)
Headache*	-	5 (4.9)	7 (7.4)
TREATMENT-RELATED AES (OCCURRING IN ≥3 PATIENTS)†			
Irritability*	3 (0.9)	2 (1.9)	2 (2.1)
Insomnia*	1 (0.3)	11 (10.7)	4 (4.3)
Headache*	0	3 (2.9)	4 (4.3)

*Patients with multiple events falling under the same term are counted only once in that term; †As judged by the investigator; AE, adverse event

TABLE 3. Cox regression analysis of time to relapse for interim and total ITT population

	INTERIM ITT POPULATION		TOTAL ITT POPULATION	
	Placebo (*n*=87)	Quetiapine XR (*n*=84)	Placebo (*n*=103)	Quetiapine XR (*n*=94)
Number of relapses (%)	36 (41.4)	9 (10.7)	50 (48.5)	11 (11.7)
COMPARISON BETWEEN TREATMENT GROUPS				
HR	0.16		0.13	
95% CI	0.08, 0.34		0.07, 0.26	
p-value	<0.0001		<0.0001	

ITT, intent-to-treat; HR, hazard ratio; CI, confidence interval

patients (82.9%) were still in the study (Figure 2). None of the patients had completed the maximum randomized treatment phase (1 year) at the time of study termination.

The treatment groups were generally well matched for demographic and baseline disease characteristics (Table 1) and no major discrepancies were noted between the populations in the stabilization and randomized phase. However, minor variations between the randomized placebo and quetiapine XR populations were observed in terms of mean age (33 and 37 years, respectively), mean disease duration (8.3 and 9.1 years, respectively) and mean number of schizophrenic episodes (3.8 and 4.9, respectively). The study population was representative of a clinically stable population of patients with schizophrenia and with minimal symptomatology (as defined by low PANSS and CGI-S scores). Patient disease characteristics were also similar in both phases of the study, although the mean PANSS total score and the mean CGI-S score were slightly higher at the time of enrollment in the open-label stabilization population compared with the population at the start of the randomized phase (Table 1).

Treatment. During the 16-week stabilization phase, the mean once-daily dose of quetiapine XR was 646mg/day (range 300–794mg/day). The most common median doses were 800 and 600mg/day, taken by 45.0 percent and 36.1 percent of the patients, respectively. The mean once-daily dose of quetiapine XR at baseline (randomization) was 649mg/day for patients assigned to the placebo group and 674mg/day for patients assigned to the quetiapine XR group. During the randomized, double-blind phase, patients in the quetiapine XR group received a mean once-daily dose of 669mg/day. The most common median daily doses were 800 and 600mg/day (48.9% and 35.1%, respectively), which demonstrates that the daily dose of quetiapine XR

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was the same in the stabilization phase as in the randomized phase. Adherence to medication (placebo or quetiapine XR) was approximately 95 percent during both phases of the study. The proportion of patients taking lorazepam, sleep medication, or anticholinergic drugs was low during both the stabilization and randomized phases, with the highest usage noted for anticholinergics and sleep medication during the initial two weeks of stabilization.

The study was terminated after a positive interim analysis; therefore, the mean duration of the randomized phase with quetiapine XR was four months (120 days) and the maximum period was nine months (270 days). During the whole study, including both the stabilization and randomization phase, 63 patients were treated with quetiapine XR for more than six months.

Stabilization phase. Patients eligible at enrollment were minimally symptomatic. Switching from previous antipsychotic medication(s) to quetiapine XR was accomplished in the first four days of the stabilization phase (Figure 1). Patients remained stable on monotherapy with quetiapine XR for the duration of the 16-week stabilization phase, as demonstrated by mean (SD) PANSS total scores of 52.60 (7.65) at enrollment, 52.49 (7.15) at Week 8 of the stabilization phase, and 52.33 (7.64) at randomization.

From the overall population who started open-label treatment with quetiapine XR (n=327), 130 patients (39.7%) discontinued treatment prior to randomization; 67 patients (20.4%) discontinued owing to the study being terminated early. Of the other 63 patients (19.3%) that discontinued during the stabilization phase the most commonly cited reason was 'not willing to continue' (35 patients [10.7%]). The available efficacy data (PANSS, CGI-S, or CGI-I) for these patients did not indicate that their psychiatric condition had deteriorated and no patients discontinued owing to lack of efficacy. Likewise, the available AE data (16 patients with no AEs and 19 patients with ≥1 AE) indicated that safety was not the reason for their withdrawal from the study. During this phase, four patients discontinued due to an AE (Table 2), of which one occurred in the first two weeks.

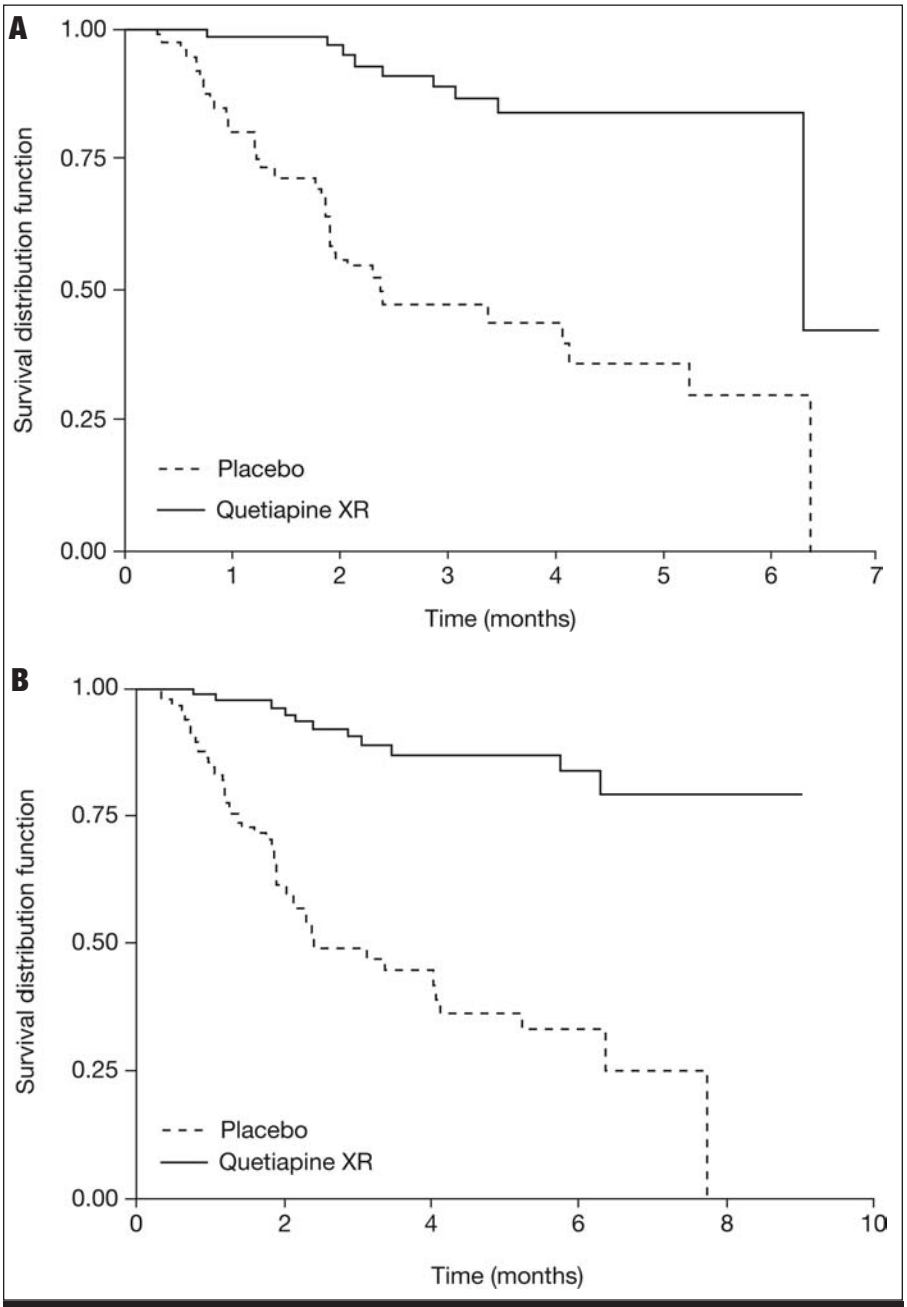


FIGURE 3. A: Kaplan-Meier curves of estimated time to relapse, interim intent-to-treat population; B: Kaplan-Meier curves of estimated time to relapse, total intent-to-treat population

Randomized phase. *Primary endpoint.* The study was terminated at the recommendation of the DSMB after the interim analysis at 45 relapses showed the difference between quetiapine XR and placebo was statistically significant in the primary outcome variable. The primary endpoint, time to first psychiatric relapse after randomization, is shown for the interim and total ITT population in Figures 3A and 3B, respectively. The low rate of relapse in the quetiapine XR group

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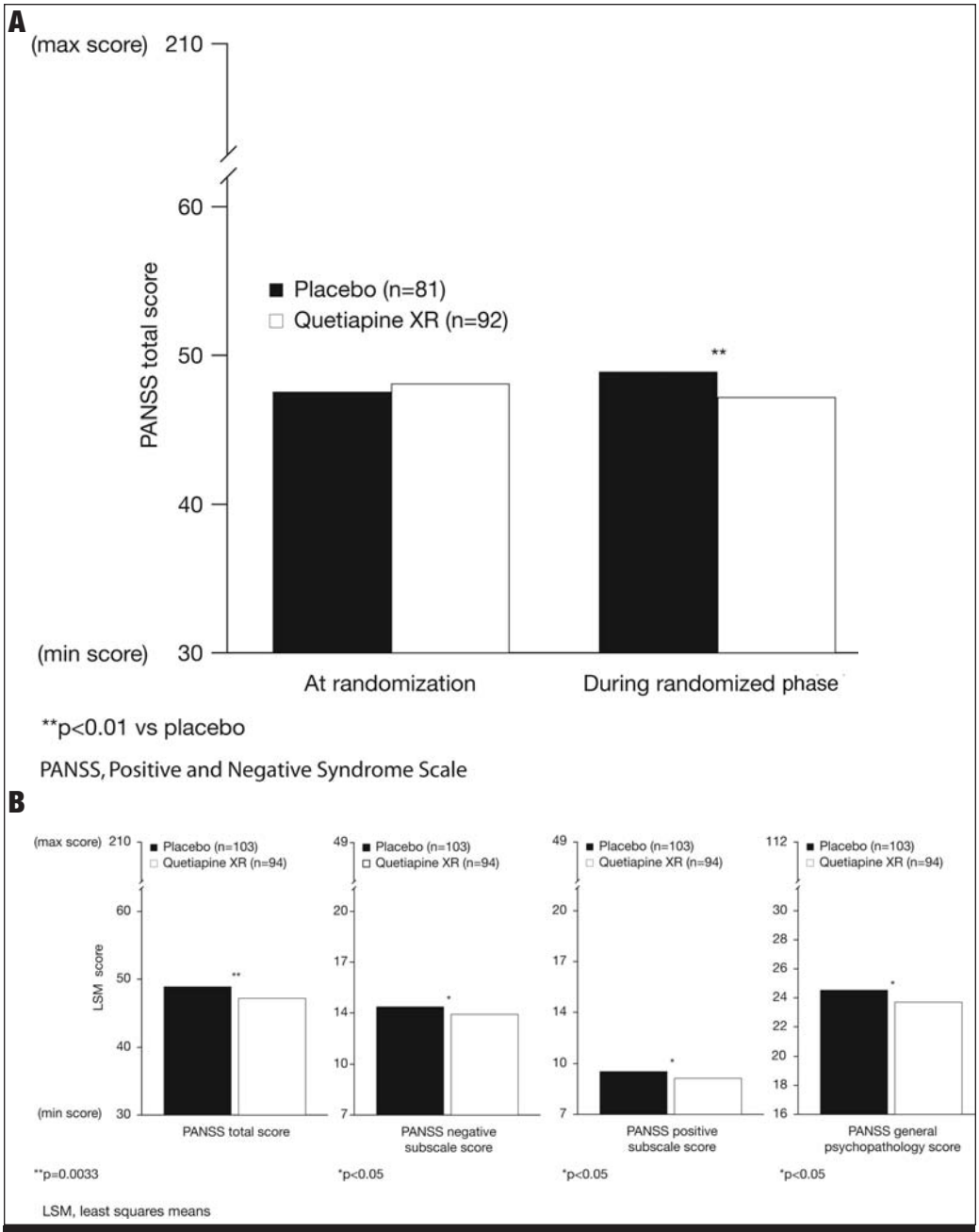


FIGURE 4. A: Mean PANSS total score at randomization and LSM PANSS total score during the randomized phase, not including relapse visit (total intent-to-treat population); B: LSM PANSS total and subscale scores at end of treatment not including relapse visit (total intent-to-treat population)

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meant that it was not possible to calculate a reliable median time to relapse. It is important to note that, in Figure 3A, the final steps after six months in the Kaplan-Meier curves are due to a late relapse of a single patient on quetiapine XR at a time when only two patients were at risk and to a late relapse of a single patient in the placebo group at a time when no other patient taking placebo was at risk. Thus, after six months of exposure, the Kaplan-Meier curves depend on single events and do not give reliable estimates of the percentage of relapse-free patients.

In the interim ITT population, the risk of a relapse was reduced by 84 percent (HR 0.16, $p<0.0001$) in the quetiapine XR-treated patients compared with placebo-treated patients (Table 3). The estimated time at which 90 percent of patients remained relapse-free was 2.9 months for quetiapine XR-treated patients, compared with 0.7 months for placebo-treated patients. Fewer patients experienced a relapse in the quetiapine XR group (9 [10.7%]) compared with patients in the placebo group (36 [41.4%]).

In the total ITT population, the risk of relapse was similarly reduced by 87 percent (HR 0.13, $p<0.0001$) in quetiapine XR-treated patients compared with the placebo-treated patients (Table 3). The estimated time to relapse was longer for quetiapine XR-treated patients, with 90 percent of patients remaining relapse-free for 3.1 months compared with 0.8 months in placebo group. Moreover, it was estimated that 80 percent of quetiapine XR-treated patients would remain relapse-free for 6.3 months compared with 1.2 months for placebo-treated patients. Additionally, in this population, 50 patients in the placebo group (48.5%) and 11 patients in the quetiapine XR group (11.7%) discontinued due to relapse.

The majority of relapses met the criteria for a marked deterioration on the PANSS scale or a high rating on the CGI-I scale (83.3% and 88.8% in the placebo and quetiapine XR groups, respectively), with no patients in the quetiapine XR group requiring hospitalization.

Secondary efficacy endpoints. The risk of relapse at six months, estimated by Cox regression analysis, was significantly lower in

the quetiapine XR group (14.3%) than in the placebo group (68.2%; $p<0.0001$; total ITT population). It should also be noted that no patients withdrew from the stabilization phase owing to worsening of schizophrenia.

Despite similarly low scores at randomization, there was a statistically significant difference in favor of quetiapine XR compared with placebo in the change in LSM PANSS total score from randomization to the last visit before relapse (47.15 vs. 48.86, respectively at last visit; $p<0.01$; Figure 4A). Unsurprisingly, when the relapse visit data were included, the difference between placebo and quetiapine XR was accentuated, with a mean (SD) difference of -6.82 (1.53) in PANSS total scores ($p<0.0001$).

The results from the analyses of the three PANSS subscales were consistent with the findings in the PANSS total score, showing that the significant difference in favor of quetiapine XR in the total score was not driven by any specific subscale component. The mean PANSS subscale scores at baseline were low and similar in both treatment groups. The estimated difference in mean PANSS subscale scores between treatment groups was significantly in favor of quetiapine XR in all cases ($p<0.05$) based on LSM scores (Figure 4B).

Maintenance of treatment effect with quetiapine XR was also supported by statistically significant differences between treatment groups in mean CGI-I scores at the last visit (placebo, 4.5; quetiapine XR, 3.7; $p<0.0001$) (Figure 5). Moreover, at the last visit the proportion of patients rated as 'improved' or 'no change' (CGI-I ≤4) was 50.0 percent and 82.8 percent in the placebo and quetiapine XR groups, respectively, with an odds ratio of 4.81 (95% CI 2.47, 9.37) in favor of quetiapine XR (Figure 5). Likewise, the proportion of patients rated as moderately ill or better (CGI-S ≤4) was significantly higher in the quetiapine XR group compared with the placebo group (93.5% vs. 84.0%, respectively; $p=0.0375$) [Figure 5]. Small numerical increases were observed in the mean (SD) changes in CGI-S scores from randomization to the last visit in both the placebo (0.7 [1.0]) and quetiapine XR (0.1 [0.9]) groups (Figure 5). However, the larger mean change in the placebo group indicated a deterioration of the global clinical status of these patients compared with the quetiapine XR group.

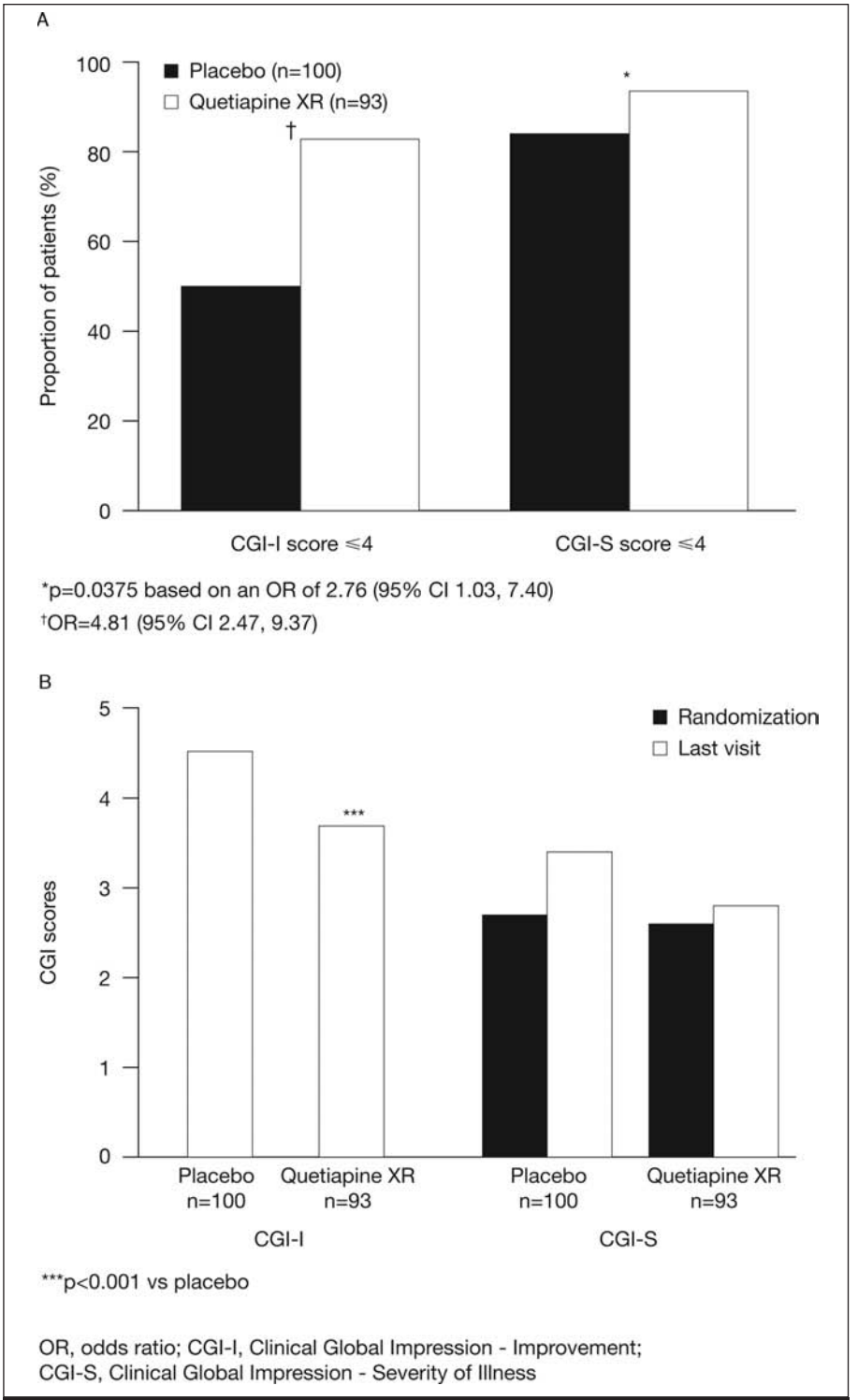


FIGURE 5. A: Proportion of patients with a CGI-I score ≤4 and a CGI-S score ≤4 at last visit (total intent-to-treat population); B: Mean CGI-I score at last visit and change in mean CGI-S score from randomization to last visit (total intent-to-treat population)

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TABLE 4. Summary of number (%) of all AEs (treatment related and unrelated) emerging during Day 1 through Day 14 after randomization

	PLACEBO (*N*=103)	QUETIAPINE XR (*N*=94)
Total AEs	29 (28)	13 (14)
Insomnia	10 (9.7)	2 (2.1)
Vomiting	3 (2.9)	0
Anxiety	2 (1.9)	0
Hyperhidrosis	2 (1.9)	0
Nausea	2 (1.9)	1 (1.1)
Left bundle branch block	1 (1.0)	0
Decreased appetite	1 (1.0)	1 (1.1)
Gastritis	1 (1.0)	0
Hyperesthesia	1 (1.0)	0
Musculoskeletal chest pain	1 (1.0)	0
Pyrexia	1 (1.0)	0
Sinus tachycardia	1 (1.0)	0
Sleep disorder	1 (1.0)	0
Stomach discomfort	1 (1.0)	0
Weight increased	1 (1.0)	0
Atrioventricular block, first degree	0	1 (1.1)
Autonomic nervous system imbalance	0	1 (1.1)
Headache	0	1 (1.1)
Influenza	0	1 (1.1)
Irritability	0	1 (1.1)
Neutrophil count decreased	0	2 (2.1)
Pain in extremity	0	1 (1.1)
Pruritus	0	1 (1.1)

AE, adverse event

Safety and tolerability. During the stabilization phase, 163 patients (49.8%) reported a total of 289 AEs; of these, 214 were considered by the investigator to be treatment related. No serious AEs were reported during this phase. The most commonly reported AEs (occurring in ≥5% of patients) during the open-label stabilization phase were somnolence and dizziness, reported by 63 (19.3%) and 21 (6.4%) patients, respectively (Table 2). The majority of the somnolence occurred during the first 14 days after enrollment (14.4%), with the frequency of emerging or ongoing somnolence reduced at randomization (2.1%) and during the randomized phase (1.9%). The frequency of all other AEs was low; three patients (0.9%) reported irritability and no other AEs were reported by more than two patients (Table 2). In the randomized phase, 22 (21.4%) and 17 (18.1%) patients reported drug-related AEs in the placebo and quetiapine XR groups, respectively. In addition, the median duration of exposure to treatment was higher for the quetiapine XR group (106 days) than the placebo group (55 days).

There was one death during the study, in a placebo-treated patient, as a result of suicide. The only other serious AE during the study was moderate 'peritonsillar abscess' in a placebo-treated patient, which was not considered by the investigator to be causally related to the study treatment. The majority of AEs reported in both the randomized and the stabilization phases were mild or moderate in intensity. During the stabilization phase, there were five AEs of severe intensity (two of somnolence and one each of constipation, dyspepsia, and epileptic seizure, all considered treatment related by the investigator). During the randomized phase, three AEs in quetiapine XR-treated patients (one each of insomnia, anxiety [considered treatment related], and contusion [not considered treatment related]) and four in placebo-treated patients were considered severe (one each of insomnia, anxiety, hyperhidrosis [considered treatment related], and the completed suicide [not considered treatment related]).

Fourteen (14%) and five (5%) patients in the placebo and quetiapine XR groups, respectively, discontinued during the randomized phase owing to reasons other than a relapse of psychiatric symptoms; of these, two patients discontinued owing to AEs (Table 2). One (1.0%) was the completed suicide in the placebo group and one patient (1.1%) discontinued owing to decreased neutrophil count in the quetiapine XR group. However, the neutrophil count of the quetiapine XR-treated patient increased from 1.26×10^9 cells/L to normal (>1.5×10^9 cells/L) while the patient was still receiving treatment.

During the stabilization phase, 11 patients had low neutrophil counts, all of which returned to normal (>1.5×10^9 cells/L) during the randomized phase. Overall, mean (SD) neutrophil counts decreased slightly during the 16-week stabilization phase (baseline, 4.49 [1.89]$\times 10^9$ cells/L;

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Week 16, 4.06 [1.70]x10^9 cells/L). In the randomized phase, four quetiapine XR-treated patients and one placebo-treated patient experienced an AE associated with neutropenia or emerging low neutrophil levels, and one patient from the quetiapine XR group withdrew because of a decreased neutrophil count. However, in all quetiapine XR-treated patients, neutrophil levels returned to normal (>1.5x10^9 cells/L) before the last visit and while continuing quetiapine XR treatment. The neutrophil count in the placebo-treated patient further decreased to 1.45x10^9 cells/L; however, the patient was discontinued owing to worsening of the psychiatric illness. During the randomized phase, small mean (SD) increases in neutrophil counts were observed in both the placebo population (randomization, 3.82 [1.37]x10^9 cells/L; last visit, 4.28 [1.78]x10^9 cells/L) and in the quetiapine XR population (randomization, 3.86 [1.67]x10^9 cells/L; last visit, 4.01 [1.71]x10^9 cells/L). No agranulocytosis was reported during either treatment phase.

Sixty-three patients (19%) experienced somnolence during the open-label stabilization phase; during the randomized phase, two patients (1.9%) receiving placebo and none receiving quetiapine XR reported somnolence. Withdrawal of quetiapine XR treatment when patients were randomized to placebo resulted in an increase in insomnia, as demonstrated by the AEs recorded for the placebo group from Day 1 through to Day 14 after randomization. No other AEs occurred in ≥5 percent of patients during this period; the total incidence of AEs was relatively low, although higher with placebo (29 [28%]) than with quetiapine XR (13 [14%]) (Table 4).

The incidence of AEs associated with EPS in the quetiapine XR group was low and comparable with placebo. This was supported by SAS scores. Baseline mean (SD) scores were 0.57 in the placebo group and 0.49 in the quetiapine XR group, with a mean (SD) change from randomization to end of treatment of -0.25 (1.12) and -0.08 (0.61) in the placebo and quetiapine XR groups, respectively. The low incidence of EPS was also reflected by the BARS scores; mean BARS scores at randomization were 0.05 in the placebo group and 0.08 in the quetiapine XR group, with a mean (SD) change to end of treatment of -0.01 (0.10) and -0.02 (0.33) in the placebo and quetiapine XR groups, respectively. Anticholinergic use was also low throughout the randomized treatment phase with no indication of increased use over time. In the placebo group, one patient was prescribed anticholinergics from randomization until Week 4. In the quetiapine XR group, two patients received anticholinergic medication until Week 12, only one patient was receiving it by Week 36, and no patient received it after Week 36.

Quetiapine XR was not associated with treatment-emergent diabetes mellitus. There was no AE associated with diabetes mellitus reported during the stabilization or randomized phase. The changes in mean and median glucose values were small and similar in the two randomized groups, and the mean changes in glucose regulation variables during the stabilization phase were small and not clinically meaningful (Table 5). Incidence of change from normal glucose levels at baseline to high levels (≥7mmol/L) at the end of treatment was 8.0 percent in the placebo-treated group and 4.1 percent in the quetiapine XR-

TABLE 5. Mean (SD) changes in documented fasting glucose, HbA_{1c}, insulin and lipid parameters from baseline to Week 16 for the open-label safety population (quetiapine XR) and from randomization to the last visit for the randomized placebo and quetiapine XR populations

	OPEN-LABEL SAFETY	RANDOMIZED, PLACEBO	RANDOMIZED, QUETIAPINE XR
Fasting glucose, mmol/L	0.05 (0.97)	0.19 (1.07)	0.34 (0.90)
	n=249	*n*=75	*n*=78
Fasting HbA_{1c}, %	0.04 (0.34)	0.03 (0.31)	0.04 (0.30)
	n=269	*n*=81	*n*=85
Fasting insulin, pmol/L	2.69 (219.80)	23.10 (111.99)	9.79 (125.32)
	n=270	*n*=81	*n*=85
Fasting HDL cholesterol, mmol/L	0.02 (0.31)	0.06 (0.27)	0.01 (0.23)
	n=272	*n*=81	*n*=84
Fasting LDL cholesterol, mmol/L	0.06 (0.68)	-0.28 (0.64)	-0.01 (0.68)
	n=267	*n*=79	*n*=83
Fasting triglycerides, mmol/L	0.05 (1.05)	-0.23 (1.31)	-0.03
	n=272	*n*=81	*n*=84

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TABLE 6. Mean (SD) changes in QTcF and standing pulse, systolic blood pressure, diastolic blood pressure from baseline to Week 16 for the open-label safety population (quetiapine XR) and from randomization to the last visit for the randomized placebo and quetiapine XR populations

	OPEN-LABEL SAFETY	RANDOMIZED, PLACEBO	RANDOMIZED, QUETIAPINE XR
QTcF	-0.95 (19.11)	-1.43 (17.85)	-3.30 (16.33)
	n=196	*n*=81	*n*=84
Pulse (bpm)	2.64 (10.07)	-1.03 (9.37)	0.60 (8.81)
	n=307	*n*=99	*n*=93
Systolic blood pressure (mmHg)	0.71 (7.76)	1.63 (7.92)	0.56 (6.39)
	n=307	*n*=99	*n*=93
Diastolic blood pressure (mmHg)	0.89 (6.93)	0.58 (6.45)	-0.91 (5.32)
	n=307	*n*=99	*n*=93

QTcF, QT interval with Fridericia correction

treated group. Mean HbA_{1c} values were relatively unchanged in both the stabilization and the randomized phases, with no median change from baseline in the stabilization phase or in either randomized group (Table 5). No patients had a shift to high-level HbA_{1c} value (>7.5%) at the end of treatment in either group. In addition, there were no clinically meaningful changes from baseline in lipid levels in the quetiapine XR group during the stabilization or randomized phases (Table 5).

During the double-blind, randomized phase, a small increase in pulse rate was observed in the quetiapine XR group; this was in contrast to a similarly small decrease in the placebo group (Table 6). There were no potentially clinically important changes in standing heart rate, systolic or diastolic blood pressure (Table 6) or in combined pulse and systolic blood pressure values (defined as ≥20 increase in pulse [bpm] and ≥20 decrease in systolic blood pressure [mmHg]) and no AEs associated with QT prolongation were reported. Moreover, no increase in mean QT interval with Fridericia correction (QTcF) was observed in either of the treatment groups at the end of the randomized phase; these findings were consistent with the stabilization phase (Table 6). There were no reports of AEs relating to orthostatic hypotension or syncope during either phase of the study; one patient (0.3%) reported tachycardia during the stabilization phase compared with two patients (2%) in the placebo group during the randomized phase.

There was a small increase in mean weight (0.89kg) during the stabilization phase, with a corresponding increase in mean waist circumference (0.35cm); 25 (8.1%) patients experienced a weight gain of ≥7 percent. Similarly, during the randomized phase, in the quetiapine XR treatment group, there was a small increase in mean weight and waist circumference (0.46kg and 0.49cm, respectively) compared with a small mean decrease (-0.68kg and -0.52cm, respectively) in the placebo group. The change in mean weight from enrollment to the end of the randomized phase in patients treated with quetiapine XR for the whole of the study was 1.06kg. The only spontaneously reported AE of increased weight was from one patient in the placebo group. Weight gain of ≥7 percent from randomization to the end of treatment only occurred in five patients (5.4%) in the quetiapine XR group and in one patient (1%) in the placebo group.

DISCUSSION

Quetiapine XR was effective in preventing relapse in patients with stable schizophrenia, as shown by the statistically significant difference in the primary endpoint in this study at the first interim analysis. Efficacy was also shown by the estimation of risk of relapse at six months, with 68.2 percent of patients predicted to relapse in the placebo group compared with 14.3 percent in the quetiapine XR group. Secondary outcome variables also supported the primary endpoint efficacy data, with patients receiving quetiapine XR experiencing small, but significant, improvements in mean PANSS total and CGI-I scores from randomization to last visit compared with those receiving placebo. While this may have been due to treatment effect, it is also likely to encompass the positive impact upon patients of more frequent monitoring and attention to adherence.

The most frequent median dose of quetiapine XR was 800mg/day, followed by 600mg/day. There was continuity in mean dose from the stabilization phase to the randomized phase, indicating that, once blinded, physicians did not increase the dose for patients in the quetiapine XR group.

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Quetiapine XR was well tolerated during the open-label stabilization and randomized phases. It was noted that a higher proportion of patients receiving quetiapine XR reported AEs during the stabilization phase compared with the randomized phase. This may have been due to AEs occurring early in the stabilization phase and resolving prior to randomization, either spontaneously or following dose adjustment. During the randomized phase, a similar number of AEs were reported by patients receiving quetiapine XR and those receiving placebo, and there were no time dependent changes in the occurrence of adverse events throughout the randomized treatment phase. Similarly, there were low levels of withdrawals and an incidence of AEs (most of which were mild to moderate) no higher than the placebo group during the randomized phase. There were no shifts outside the normal ranges in metabolic parameters during this study and the majority of weight gain occurred during the 16-week stabilization phase (0.89kg of a total mean gain of 1.06kg). The proportion of patients reporting somnolence as an AE decreased from the stabilization phase to the randomized phase, indicating that this AE generally does not persist during longer-term treatment with quetiapine XR. Similarly, the incidence of EPS-related AEs was low during the randomized phase. Compared with quetiapine IR, there were no new safety or tolerability concerns associated with quetiapine XR treatment.[19]

This study design included a 16-week stabilization phase with a four-day cross-tapering approach to switch from previous antipsychotic to quetiapine XR. Patients eligible at enrollment were minimally symptomatic, with low PANSS and CGI scores. Even with this fairly rapid switch to quetiapine XR, withdrawals were low (excluding the early cessation of the study). This provides preliminary evidence that patients can be switched from their previous medication over four days without difficulty and supports previously published data for quetiapine IR.[23] In addition, all patients remained symptomatically stable on monotherapy with quetiapine XR during this 16-week stabilization phase up to randomization, as demonstrated by the PANSS scores. Although 49.8 percent of patients in the stabilization phase experienced AEs, only four out of 327 patients were withdrawn due to AEs, suggesting that quetiapine XR was generally well tolerated. Withdrawal due to patient decision at this stage could have been associated with the patient's fear of being taken off an active and effective treatment and being placed into the placebo arm of the study, a phenomenon that has been documented previously.[24,25]

High relapse rates, attributed to a rebound effect, have previously been shown for chlorpromazine, haloperidol, and other typical antipsychotics after the abrupt cessation of treatment.[26] The types of AEs associated with the withdrawal effect of antipsychotics such as chlorpromazine include nausea, vomiting, headaches, diarrhea, perspiration, restlessness, and sleeplessness, all of which are usually transient and occur in the first two weeks.[27–29] In this study, quetiapine XR was withdrawn from patients randomized to the placebo arm relatively abruptly (4-day cross titration; Figure 1); however, no early rebound effect was apparent, as indicated by the fact that relapses did not occur more frequently during the first few weeks after randomization (Figure 3). Moreover, during the first two weeks after randomization, 28 percent of patients switched to placebo reported an AE compared with 14 percent of patients receiving quetiapine XR (Table 4). Of these, 9.7 percent reported insomnia, 1.9 percent reported anxiety and 1.9 percent and 2.9 percent reported nausea and vomiting, respectively, and none reported headaches or restlessness. The relatively high incidence of insomnia reported by patients switched from quetiapine XR to placebo may be a consequence of the removal of histamine receptor blockade.[30] These data suggest that, other than insomnia, the withdrawal of quetiapine XR is relatively well tolerated.

As with all clinical trials, this study has some limitations. In this case, interpretation is limited because patient choice (which could represent lack of efficacy or tolerability issues) was the reason for the majority of the withdrawals during the stabilization phase. Also, this study was terminated early when there was only a four-month mean duration of randomized treatment. However, this study is unique in that the analysis of relapse was not restricted to patients who had demonstrated a short-term treatment response. Inclusion criteria for this study required patients to be minimally symptomatic or clinically stable prior to randomization, which is comparable with that of a relapse prevention study of olanzapine in patients with schizophrenia.[31] As such, these trials have determined the efficacy of an antipsychotic to maintain symptom control and in this context a *relapse* could also be defined as a *new psychotic episode*. In contrast, several placebo-controlled studies investigating relapse rates may be more appropriately described as *continuation* studies.[32,33] In these trials, patients were enrolled with an acute psychotic episode and were randomized following a positive outcome in a short-term efficacy study; therefore, worsening of schizophrenic symptoms in this setting may be a true relapse of the recent psychotic event. Studies looking at continuation phase rather than maintenance phase limit the extrapolation of these data to routine clinical practice.[1] Consequently, it can be difficult to compare data across these studies, which have different periods of

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KEY POINTS

- Extended release quetiapine fumarate (quetiapine XR) is a new formulation that allows convenient once-daily dosing.
- Quetiapine XR is significantly more effective than placebo at preventing relapse in patients with schizophrenia.
- Quetiapine XR was well tolerated during longer-term treatment, with low levels of discontinuations and AEs.

stabilization preceding randomization and are of varied duration and design.

Another advantage of this design, as compared with a design employing a double-blind stabilization phase followed by an extension phase, is that patients responding to placebo are not included in the maintenance phase. The manner in which the data were analyzed in this study was robust, utilizing a five stepwise sequential procedure for the confirmatory part of this study to ensure a multiple level of significance of 0.05. Previous placebo-controlled studies examining the medium- to long-term efficacy of ziprasidone, olanzapine, amisulpride, and zotepine have all utilized two-tailed analyses.[32–35] While a two-tailed analysis is an accepted and established measure of significance, the five stepwise approach controls for the possibility of false positive results more effectively.

Although it is difficult to draw comparisons owing to the variability of study designs as noted above, it is of interest to consider relapse rates reported in other long-term studies investigating atypical antipsychotics. Two trials examining relapse prevention of risperidone reported relapse rates of 30 percent and 34 percent after one year of treatment.[36,37] In a 26-week, placebo-controlled, randomized, double-blind study of aripiprazole in patients with schizophrenia experiencing ongoing stable symptomatology, the relapse rate was 34 percent (significantly lower than with placebo; 57%, $p<0.001$); however, the patients enrolled in this study had a relatively high baseline PANSS total score of 82 points.[38] Reports on the relapse prevention efficacy of olanzapine vary, with rates from four trials ranging from 27 to 54 percent after one year of treatment.[1,39] In addition, one olanzapine trial designed to assess relapse in clinically stable patients demonstrated superiority to placebo ($p<0.0001$), with estimated relapse rates at six months of 5.5 percent for olanzapine-treated patients and 55.2 percent for placebo-treated patients.[31] An open-label, naturalistic, head-to-head study comparing olanzapine, risperidone, and quetiapine IR has reported relatively low relapse rates of 7.7 percent, 9.0 percent, and 12.5 percent, respectively, after one year of treatment; however, patients were selected on the basis of adherence to treatment.[40] In this study, the relapse rates during the four-month mean treatment period with quetiapine XR were 10.7 percent and 41.4 percent with placebo, with an estimated six-month relapse risk of 14.3 percent for quetiapine XR and 68.2 percent for placebo.

An important consideration for future relapse trials is that although the use of a placebo arm is important to demonstrate efficacy or lack of efficacy for any treatment, its use in schizophrenia clinical trials is being questioned owing to the severe nature of the condition and the considerable negative impact of relapse on illness severity and future treatment.[3] However, it is notable that the regulatory authorities in the countries in which this study was conducted required the use of a placebo in this study to determine the absolute efficacy of quetiapine XR in reducing the risk of relapse. In the present study, the risk to patients was minimized by the DSMB conducting pre-planned interim analyses after 45 and 60 observed relapses, as well as having persons monitoring the safety of the patients to enhance clinical care during this study. For future studies, it may, therefore, be appropriate to investigate the differences between formulations of the same antipsychotic or versus an active comparator.

Overall, the design of this study allows us to address the question of whether continued therapy on quetiapine XR is necessary for preventing relapse. Although there is no precise definition of relapse, the relapse criteria developed for this study are largely consistent with those used in other studies[32,35,36,38] and encompass clinically relevant changes in the status of a patient. Therefore, these results show that quetiapine XR is suitable as a once-daily medication for the long-term treatment of schizophrenia, and that it is both well tolerated and effective in preventing relapse.

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